When Schema Lexicons Fail: Diagnosing and Repairing Entity-Layer Collapse in a Production Knowledge Base

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ABSTRACT

In dictionary-driven entity enrichment, the schema lexicon *is* the extraction model. We report a production incident in which a category-word lexicon silently collapsed the entity layer of a 61,137-document scholarly knowledge base: 575 type-level entities ("method", "energy") carrying ~19.2M links while every job-level monitor stayed green. We diagnose the failure across three layers (extraction mechanism, lexicon provenance, monitoring blindness), repair it with a three-source corpus-presence-attested lexicon (3,702 terms), rebuild the entity layer under production load, and derive scale-invariant quality monitors. An isolated fixed-code, fixed-corpus experiment that varies *only the lexicon* reproduces the collapse signature alone (77 entities, top-10 concentration 0.557, 1.8 concepts/document) and isolates the repair effect (493 entities, 0.316, 3.7); the rebuilt layer carries 1,871 entities, 25.1M links, and a functioning relation layer; distinct concepts per document rise 15.7→37.8 with document coverage 48.5%→98.5%; an LLM-assisted audit puts link precision at 0.85 (95% CI [0.77, 0.91]; 1.00 on domain terms, n=25). The generic tail survives repair (9% of links sit on the single ambiguous symbol "be"), retrieval-facing gains are not attributable to the lexicon alone (entity-channel ablation bitwise null), audit is model-assisted rather than human-adjudicated, and monitor thresholds are post-hoc on 13 libraries — limitations stated up front.

KEYWORDS

knowledge base construction schema quality terminology extraction entity linking production incident observability

1 INTRODUCTION

Scholarly knowledge platforms increasingly generate subject schemas with LLM assistance: a batch tool drafts entity types with a handful of aliases each, a validator checks structural well-formedness, and the schema is bound to knowledge bases as configuration [5]. When the enrichment pipeline then extracts mentions by *dictionary scan*, the schema's alias list becomes the extraction surface — the lexicon is not descriptive metadata but operative infrastructure.

The failure we report: a 61,137-document nuclear-electronics knowledge base accumulated ~19.2M entity links onto **575 canonical entities**, every one a bare type word ("method", "parameter", "energy"); a 13,408-document physics library showed the same disease at 2,018 entities. Every operational signal stayed green — jobs done, throughput nominal, link volume reading as growth. The graph had degenerated into a bipartite type–document index that was row-level indistinguishable from a populated knowledge graph.

We contribute: (1) a forensic record of schema-lexicon-induced entity collapse, with the timeline and the observables that expose it (top-10 link concentration and the top-entity name audit; cardinality only corroborates); (2) a three-source corpus-presence-attested lexicon construction method (seed transplant, corpus-candidate classification, bounded expansion) that repaired both schemas without manual lexicography; (3) an *isolated* fixed-code fixed-corpus experiment showing the lexicon variable alone reproduces the collapse and its repair; (4) a full-chain production rebuild whose forensics corrected our own incident log (an `INSERT IGNORE` mis-attribution mechanism and a seven-hour "zero growth" claim contradicted by row timestamps); and (5) four gold-free quality monitors with a 13-library calibration.

Table 1: The collapse, measured, with snapshot provenance ("bckp." = 2026-10-06 cold backup, mid-enrichment; "disc." = discovery-day).

KB	Entities	Links
nuclear (61,137 docs)	**575** (disc.) / 503 (bckp.)	~**19.2M** (disc.) / 4.81M (bckp.)
physics (13,408 docs)	**2,018**	2.23M (bckp.)
hydro (ref., 78 docs)	1,608	719,515

2 BACKGROUND AND PRIOR ART

Dictionary- or gazetteer-driven entity recognition is one of the oldest NER traditions [2]; corpus-statistical term recognition goes back to C-value/NC-value [1] — our corpus-candidate source is that program reborn with an LLM classifier. Knowledge-graph refinement [3] and production data validation [4] address adjacent problems. What prior art does not cover is the specific failure locus: when the dictionary *is* the extraction model, a structurally valid but terminologically hollow schema passes every existing gate.

Our pipeline extracts mentions in stage K1 by normalized dictionary scan (CJK substring; ASCII word-boundary with optional plural, case-folded); K2 links mentions to canonical entities by normalized form + type; K3 adjudicates typed relations with an LLM; K4 projects to the graph store. The healthy reference case in the same deployment — hydrogeology, 78 documents, a hand-evolved 3,742-term lexicon, 1,608 entities — establishes plausibility, not causation: domain, scale, and provenance all differ.

3 THE COLLAPSE, FORENSICALLY

Observation (Table 1). During an unrelated storage investigation (2026-10-08), a routine entity-quality sample returned type words where domain concepts were expected.

Mechanism. When the alias set for type *method* contains common words, every occurrence in 7.2M chunks becomes a mention, and K2 folds them onto one entity per (form, type). The mechanism executed faithfully on degenerate input; the dictionaries were the schemas — the batch generator produced a category taxonomy where a terminology was needed [6].

Timeline (compressed): 10-08 discovery → root cause, lexicon tool built/run/committed, teardown + 766,501-event re-enqueue; 10-08 night: silent mis-attribution window (below); 10-09 drain completes, statistics recomputed, two smaller diseased libraries treated; 10-09/10: retrospective audits (scoped refresh, channel ablation, link audit, fleet-wide concentration) and manuscript. The compression is real; the evaluation is retrospective, not pre-registered.

Why nothing fired. Job-level observability was all green; link volume read as success. The observables that expose the collapse are the concentration of links on the top entities (collapsed: 0.419, measured from the pre-incident backup, versus 0.18 for the reference case) and the name distribution of those entities (pure type vocabulary). Entity cardinality per document — a factor of 2,192 below the reference case (0.0094 vs 20.6) — corroborates, but is gameable in both directions, so it is the secondary signal.

4 CORPUS-ATTESTED LEXICON CONSTRUCTION

Three sources, all gated by corpus presence (a term the corpus never attests is not admitted — presence is necessary, not sufficient, for termhood): **seed transplant** (2,284 terms from a healthy near-domain library); **corpus-candidate classification** (top 8,000 unigram/bigram n-grams by chunk frequency over a 3,000-chunk-per-KB sample, frequency ≥5 after boilerplate and stop-word filtering; an LLM classifies each into a schema type or rejects it); **per-type expansion** (≤60 per type). Catch-all types accept no classified noise; cross-type conflicts resolve by specificity. Classifier: Qwen3.6-35B-A3B (AWQ 4-bit, community build; served under a historical alias), temperature 0.2, single idempotent run. Output: nuclear v2.1 = 3,702 terms; physics v1.1 = 1,279.

5 REBUILD AT SCALE — AND A CORRECTED INCIDENT LOG

Per knowledge base: delete graph vertices with edges, the four relational tables in bounded chunks, and the search-side projection; re-enqueue all documents through four stages. The printed "766,501 jobs" counts *events*, not distinct doc-stages (74,545×4 = 298,180): the surplus is the repair window's re-enqueue waves (silent-empty reset 75,831; zero-effect relations/graph resets ≈149k; watchdog-deletion recovery ≈122k; small libraries 5,772).

Four pipeline pathologies surfaced and were fixed (committed, measured): an unbounded projection join that saturated a 112-core database at load 500 (scoped to per-document mentions: 25+ min → seconds); two link-insertion deadlock classes (deadlock 1213 → 0); and an $O(N^2)$ alias check the richer lexicon itself amplified (101 s → 0.05 s per task; fleet link throughput 8/min → 1,100/min — a window-level aggregate, with the cache's isolated contribution measured at task level).

The fifth pathology, revisited. Our log recorded a parameter-order bug in a single-statement link writer whose rows `INSERT IGNORE` "silently swallowed" for seven hours. A minimal reproduction on the production engine (MySQL 8.4, `STRICT_TRANS_TABLES`, faithful DDL) sharpens the mechanism: the NULL from `UNHEX('alloy')` is **replaced by the implicit default 0x00×16 and the row is inserted** — into a ghost bucket invisible to every per-KB query — and a later correct re-insert for the same (mention, entity) pair is **silently skipped by the unique key**: the orphan poisons the repair unless deleted first (production carries zero 0x00 rows today, so cleanup happened). The duration claim also fails row-level forensics: the registry's own `created_at` histogram (Figure 1) shows KB1 inserting at 1.0–2.3M rows/hour *through* the nominal silent window and KB2's entire link layer landing in one hour. We keep the log entry and its correction side by side: incident narratives written from job dashboards are themselves unverified claims.

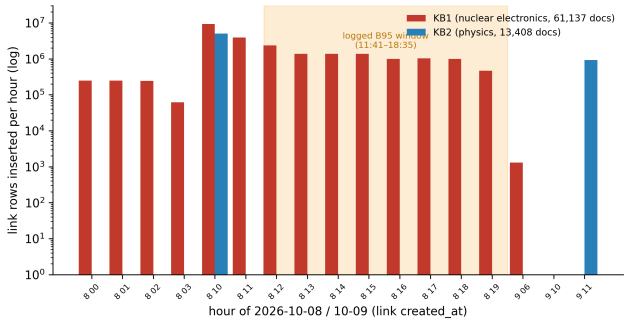


Figure 1: Rebuild telemetry from the link registry's own timestamps. The shaded band is the *logged* silent window; inserts continue through it — the timestamp correction of §5.

Table 2: Entity layer before/after, with snapshot provenance.

Metric (KB1)	Collapsed	Repaired
canonical entities	575 / 503 (10-06)	**1,871**
entity links	~19.2M / 4.81M	**25.1M**
typed relations	~0	**44,043**
top-10 concentration	0.419	**0.345**
concepts/doc (all-doc)	15.7 (48.5% empty)	**37.8** (98.5%)
KB2: concepts/doc	29.5	**61.4**

6 BEFORE/AFTER

Concepts-per-document replaces the raw entity/document ratio: its numerator scales with documents and the coverage column exposes the collapse's most user-visible symptom — half the library had *no* annotations. Absolute link counts *rise* after repair (KB1 19.2M→25.1M; KB2 2.23M→5.0M): real terms match more often. Count was never the quality signal; distribution and names are.

Name quality. The repaired head is still generic: `be` (2.26M links, 9.0%), `alpha`, `gamma`, `beta`, `method` (Figure 2). The `be` case is a normalization failure — element symbol and auxiliary verb merge under case-folded matching — and §6.1 prices its treatment. The repair moved the problem from "no terminology at all" to "terminology plus a generic tail."

Retrieval: honest attribution. Scoped coverage rose +6.6/+7.3 points over the repair window, but three controls bound the attribution: (a) a single-variable channel ablation under the final configuration is *bitwise null* (removing all entity channels changes no

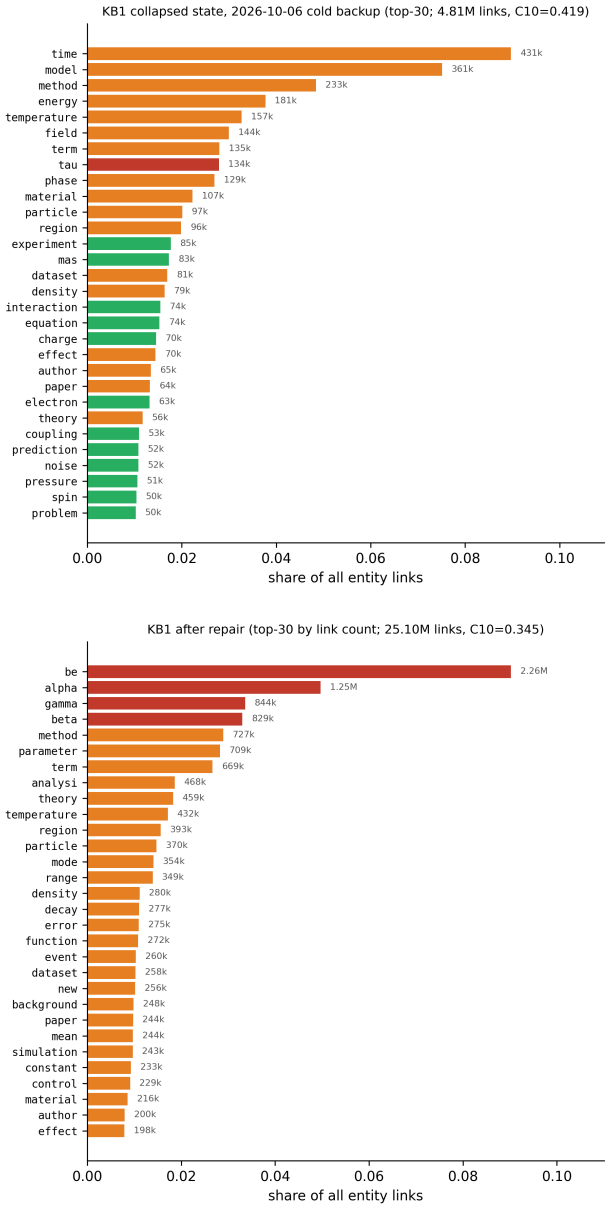


Figure 2: Top-30 entities by link share, before (cold backup; C_{10}=0.419) and after repair (C_{10}=0.345). Green = domain terms, orange = generic words, red = short ambiguous symbols.

metric while halving latency; a forced-trigger check on the natural-phrasing set shows the channels are alive — 54/99 fired, and killing them costs 20 losses / 0 wins); (b) alias expansion, the one entity-adjacent mechanism, is +1.5pt (ns) on the benchmarks at 2× latency; (c) two same-configuration runs differ on exactly one query (0.5pt noise floor). The trajectory is system-level; the entity layer's retrieval-facing value runs through asset surfaces (documented in the companion system manuscript [7]), not fusion votes.

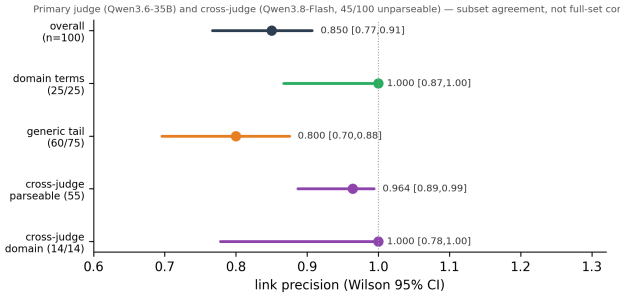


Figure 3: Audit precision with Wilson 95% CIs; cross-judge bars apply to the 55-link parseable subset only.

Table 3: Isolated lexicon comparison (fixed code, fixed corpus).

Arm	Links	Ents	C_{10}	Conc/doc
v2.0 collapse-era	2,720	77	**0.557**	1.8
v2.5 mid	4,472	266	0.339	2.4
v2.1 repair	9,456	493	0.316	**3.7**
+case-sens. short	8,328	468	0.290	3.4
+stop-word block	8,756	492	**0.276**	3.5

Link precision audit. A seeded 100-link sample (span integrity 100/100) judged by strict rubric: precision **0.85** (Wilson CI [0.77, 0.91]); domain stratum 25/25 (n=25 — small); generic stratum 60/75. A second Qwen-series model re-adjudicated: 45/100 responses unparseable (flash-tier), and on the 55-link parseable subset agreement is 96.4% (domain 14/14) — a cross-model check on a subset, not family-independent confirmation (Figure 3). Human adjudication remains the decisive future control.

6.1 The Lexicon Variable, Isolated

The repair bundled lexicon, teardown, and fleet changes; this experiment isolates the lexicon alone: a semantic replica of the K1/K2 scan-and-link semantics over a deterministic 2,000-chunk sample, with *only the dictionary swapped* — v2.0 as committed at library creation (182 scan terms), v2.5 mid-evolution (2,932), v2.1 the repair (3,733), plus two v2.1 governance variants.

Three findings (Table 3). **(1)** The collapse signature is reproduced by the lexicon *alone*: 77 entities, C_{10}=0.557, 1.8 concepts/document — same code, same corpus. **(2)** The repair effect is isolated: ×6.4 entities, ×2.1 concepts/doc, generic-head mass 0.163→0.062; the production collapsed state (503 entities, C_{10}=0.419) sits between the v2.0 and v2.5 counterfactuals, consistent with mid-enrichment dictionary evolution. **(3)** The be class is treatable at known cost: case-sensitive short symbols remove the largest entity (7.4% of links; be leaves the top-5; C_{10} →0.290) at a fold-recall price (links −11.8%, judged precision 0.95→0.925); a stop-word block is gentler (−7.4%, 0.276, 0.975). Judged precision is near-tautological for dictionary self-matches, so structural metrics lead. *Provenance*: decomposing v2.1's link mass by which committed lexicon first contained each

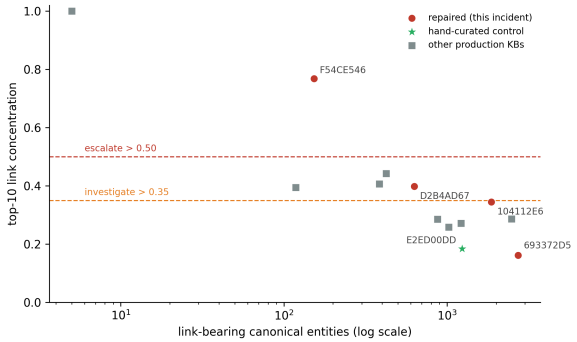


Figure 4: Top-10 link concentration vs. link-bearing entities across 13 production libraries; red = rebuilt in this incident, green star = hand-curated reference. Bands are the proposed post-hoc thresholds.

term — v2.0-inherited 21.4%, mid-evolution 15.3%, **three-source-new 63.2%** (296 of 493 entities); a true single-source regeneration ablation remains future work.

7 MONITORS AND DISCUSSION

Four gold-free monitors, in the spirit of production data validation [4]: **(a) top-10 link concentration** (primary, scale-invariant): across the 13 deployment libraries the metric spans 0.16–0.77; healthy/repaired large libraries run 0.16–0.19, the rebuilt nuclear library 0.345, the two small treated libraries *currently* 0.398 and 0.768 (reported as measured — the smallest also shows the metric's Zipfian floor effect); collapsed state measured 0.419. Proposed bands (>0.35 investigate, >0.5 escalate) are post-hoc on these 13 libraries — a starting point, not a validated norm; prospective validation is future work. **(b) entity cardinality ratio** (coarse secondary, gameable both ways). **(c) top-entity name audit** (five-minute quarterly control). **(d) target-table growth reconciliation** — the B95 ghost-bucket mechanism makes this mandatory wherever a bulk writer is guarded by `INSERT IGNORE`. The small-library treatment doubles as out-of-sample validation: cardinality normalized (870→629, 209→153) while concentration still flags both — the two monitors separate "cardinality repaired" from "distribution still skewed."

The incident is not an argument for LLM-everywhere: if extraction is a scan, the dictionary *is* the model and deserves a trained model's evaluation discipline. Admission criteria (corpus presence) are necessary but not sufficient; *ranking* criteria (IDF-style contrast, bilingual consistency, length/type priors) are the open problem — the generic tail is the standing evidence.

8 LIMITATIONS

(1) The generic tail survives repair (9% of links on "be"). (2) Retrieval attribution is bounded, not eliminated: the full lexicon×configuration decomposition would require re-collapsing a production library. (3) Coverage: two libraries rebuilt, two smaller treated; bilingual anomalies documented, not explained. (4) Six secondary incidents extended the repair; reported as findings. (5) Audit and cross-judge are model-assisted (same vendor family), not human-adjudicated; a stratified human sample is the next evaluation step. (6) The isolated experiment uses a semantic-replica scanner and committed-snapshot dictionaries, not independently regenerated single-source lexicons.

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